1.8.2 Simplify each of the following without a calculator:

(b) $\sqrt[4]{81}$

1.8.6 Find the integers a and b such that $\dfrac{\sqrt{600}-\sqrt{150}+3\sqrt{54}}{6\sqrt{32}-2\sqrt{50}-\sqrt{288}} = a\sqrt{b}$, and $\sqrt{b}$ cannot be simplified.

2.41★ Alice, Bob, and Carol each think of an expression that is a fraction with 1 as the numerator and a constant integer times some power of x as the denominator. The simplest common denominator of Alice's and Bob's expressions is $4x^2$. The simplest common denominator of Bob's and Carol's expressions is $12x^3$. The simplest common denominator of Alice and Carol's expressions is $6x^3$.

Find all possible expressions that could be Carol's expression. **Hints:** 49

4.34 Simplify the fraction $\dfrac{\dfrac{3r^2t - 6rt^2}{6r^2t^3}}{\dfrac{8t^3r - 4t^2r^2}{6r^3t^4}}$.

5.4.3 The sum of Eric's and Bob's weights is 9 times the difference of their weights. The positive difference of their weights is also 240 pounds less than the sum. If Eric weighs less than Bob, find Bob's weight.

5.6.3 In the magic square shown, the sum of the numbers in each row, column, and diagonal are the same. Five of these numbers are represented by v, w, x, y, and z. Find $y + z$. *(Source: AMC 12)*

v	24	w
18	x	y
25	z	21

5.25 For what value of the constant c does the system of equations below have no solutions (x, y)?

$$3x - 5y = -2.3,$$
$$6x = cy + 9.3.$$

5.30 Solve the system of equations below:

$$\begin{aligned} p + 3q - 2r &= 19, \\ -2p + 7q - 6r &= 26, \\ -p - 2q + 8r &= -31. \end{aligned}$$

5.36 Find all ordered pairs (x, y) that satisfy both $\dfrac{3x - 4y}{xy} = -8$ and $\dfrac{2x + 7y}{xy} = 43$. **Hints:** 20

5.38★ One morning each member of Angela's family drank an 8-ounce mixture of coffee with milk. The amounts of coffee and milk varied from cup to cup, but were never zero. Angela drank a quarter of the total amount of milk and a sixth of the total amount of coffee. How many people are in the family?

6.5.7 The state income tax where Kristin lives is charged at the rate of $p\%$ of the first \$28000 of annual income plus $(p+2)\%$ of any amount above \$28000. Kristin noticed that the state income tax she paid amounted to $(p+0.25)\%$ of her annual income. What was her annual income? *(Source: AMC 12)*

6.48 Given that $yz : zx : xy = 1 : 2 : 3$ and $\frac{x}{yz} : \frac{y}{zx} = 1 : k$, find k. *(Source: AHSME)* **Hints:** 40

6.52★ Cassandra sets her watch to the correct time at noon. At the actual time of 1:00 PM, she notices that her watch reads 12:57 and 36 seconds. Assuming that her watch loses time at a constant rate, what will be the actual time when her watch first reads 10:00 PM? *(Source: AMC 12)* **Hints:** 61

6.55★ It is now between 10:00 and 11:00 o'clock, and six minutes from now, the minute hand of a watch will be exactly opposite the place where the hour hand was three minutes ago. What is the exact time now? *(Source: AHSME)* **Hints:** 102

Problem 7.15: Two trains, each moving at 20 miles per hour towards each other, are initially 60 miles apart. A bee starts at the front of one train, flies to the other train, then back to the first train, and so on. If the bee always flies at 30 miles per hour, how far does the bee fly before the trains collide? **Hints:** 7

7.31 Two teams compete in a relay race; they are composed of four members, each of whom runs 100 meters. All the members of team B run at b meters per second (mps). The first three members of team A can only run $5b/6$ mps, while the fourth member runs at a mps. Find a/b if the race is a tie. *(Source: Mandelbrot)*

7.32 Two sisters ascend 40-step escalators that are moving at the same speed. The older sister can only take 10 steps up the crowded "up" escalator, while the younger sister runs up the empty "down" escalator unimpeded, arriving at the top at the same time as her sister. How many steps does the younger sister take? *(Source: Mandelbrot)*

7.38 Jenna and Ginny are 20 miles from home. They have one pair of roller blades. Jenna walks 4 mph and skates 9 mph. Ginny walks 3 mph and skates 8 mph. They start for home at the same time. First, Ginny has the roller blades and Jenna walks. Ginny skates for a while, then takes the roller blades off and starts walking. When Jenna reaches the roller blades, she puts them on and starts skating. If they both start at 4:00 and arrive home at the same time, what time is it when they get home? **Hints:** 1

7.42 On an auto trip, the distance read from the instrument panel was 450 miles. With snow tires on for the return trip over the same route, the reading was 440 miles. Find, to the nearest hundredth of an inch, the increase in radius of the wheels if the original radius was 15 inches. *(Source: AMC 10)* **Hints:** 210

7.45★ In an h-meter race, Sunny is exactly d meters ahead of Windy when Sunny finishes the race. The next time they race, Sunny sportingly starts d meters behind Windy, who is at the starting line. Both runners run at the same constant speed as they did in the first race. In terms of h and d, how many meters ahead is Sunny when Sunny finishes the second race? *(Source: AHSME)* **Hints:** 29, 134

7.47★ A and B travel around a circular track at uniform speeds in opposite directions, starting from diametrically opposite points (meaning they are directly opposite each other on the track). If they start at the same time, meet first after B has traveled 100 yards, then meet a second time 60 yards before A completes one lap, then what is the circumference (length) of the track? *(Source: AHSME)* **Hints:** 9, 78